

VAE-1: Generative Modeling and Latent-Space Exploration

VAE family · CORE · 30 points

1. Problem & Data

This project trains a Variational Autoencoder (VAE) to model Fashion-MNIST, a dataset of 70,000 grayscale 28×28 clothing images across 10 classes (60,000 train / 10,000 test), obtained via torchvision. The goal is to learn a compact latent representation from which new images can be generated, existing images can be smoothly interpolated, and the geometry of the learned space can be probed directly.

2. Method

A convolutional VAE was implemented with a 3-layer convolutional encoder (32→64→128 channels) producing a 20-dimensional latent code, and a mirrored transposed-convolutional decoder. Training used the standard ELBO objective (binary cross-entropy reconstruction loss + KL divergence to a standard normal prior), Adam optimizer (lr = 1e-3), batch size 128, for 20 epochs. To avoid posterior collapse — a common failure mode where the KL term dominates before the decoder has learned anything useful — the KL weight β was linearly warmed up from 0 to 1 over the first 5 epochs rather than fixed at 1 from the start.

3. Original Contribution: Quantitative Latent-Space Attribute Manipulation

Hypothesis: Linear directions in the VAE's latent space correspond to semantically meaningful, class-level attributes. Specifically, moving a source-class latent vector along the direction $\text{mean}(\text{target_class}) - \text{mean}(\text{source_class})$ should smoothly shift the decoder's output toward the target class — and this can be measured objectively, not just eyeballed.

To test this without relying on subjective judgment, an independent CNN classifier ("oracle", not part of the VAE) was trained on Fashion-MNIST to 89.24% test accuracy. For three visually-related but distinct class pairs — Sneaker→Ankle boot, Pullover→Coat, and T-shirt/top→Shirt — the per-class mean latent vector was computed from real encoded images, and up to 200 source-class test images were shifted along the target-minus-source direction at seven increasing strengths ($\alpha = 0$ to 3.0). Each shifted latent was decoded and re-classified by the oracle; the fraction now labeled as the target class (the "flip rate") is the quantitative evidence for or against the hypothesis.

4. Results

4.1 Training Convergence

The ELBO decreased steadily on both the training and test sets with no sign of divergence (test loss tracked train loss closely throughout), indicating stable optimization. The visible bump in test KL right as β reaches 1.0 (epoch 5) is the expected transient from the end of KL warmup.

Epoch	β (KL weight)	Train ELBO	Test ELBO
1	0.20	262.29	267.39
5	1.00	243.74	244.72
10	1.00	239.88	241.54

Epoch	β (KL weight)	Train ELBO	Test ELBO
20	1.00	237.63	239.52

4.2 Reconstructions, Samples, and Interpolation

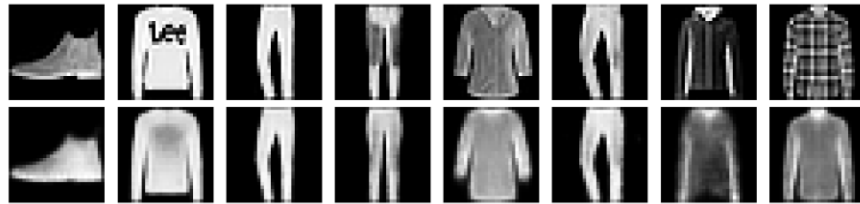


Figure 1. Top row: real test images. Bottom row: VAE reconstructions. Overall garment silhouette is preserved; fine texture and printed detail (e.g. the “Lee” logo, plaid pattern) is smoothed away, as expected from a 20-D bottleneck under BCE reconstruction loss.



Figure 2. 64 samples drawn from the prior $N(0, I)$ and decoded. Most samples are recognizable, coherent garment/shoe silhouettes, indicating the decoder has learned a well-formed generative mapping rather than memorizing training images.

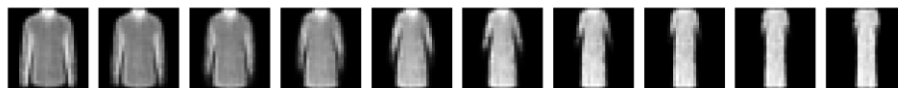


Figure 3. Linear interpolation in latent space between two real test images (10 steps). The transition passes through plausible intermediate garment shapes rather than producing blurry or incoherent frames — the qualitative signature of a well-regularized latent space (see Discussion, Q1).

4.3 Latent-Space Structure (PCA Projection)

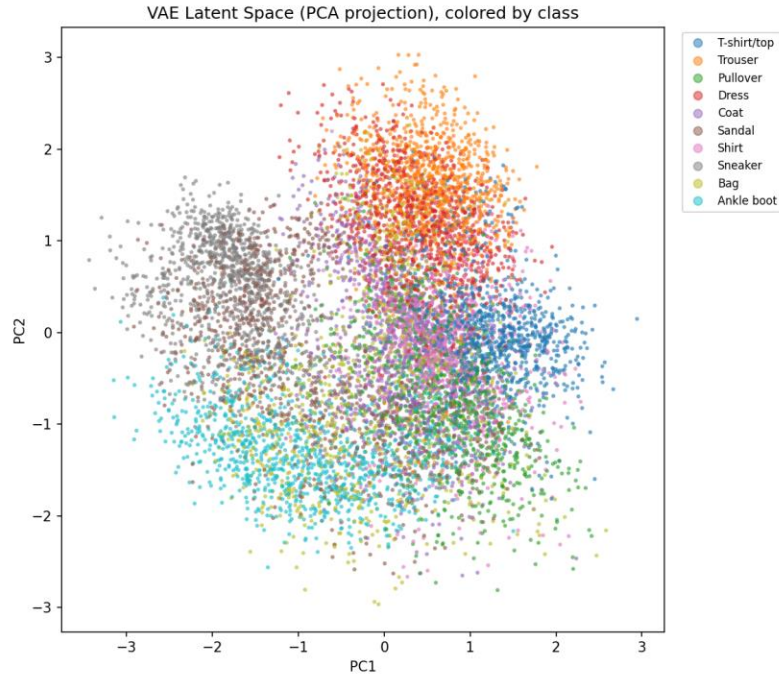


Figure 4. 2-D PCA projection of the 20-D latent space, colored by class. Footwear (Sneaker, Sandal, Ankle boot) separates cleanly from apparel, and Trouser forms its own tight cluster — both visually and structurally distinct categories. Upper-body garments (T-shirt/top, Pullover, Coat, Shirt, Dress) overlap substantially, foreshadowing the attribute-manipulation results below.

4.4 Quantitative Attribute Manipulation — Original Contribution

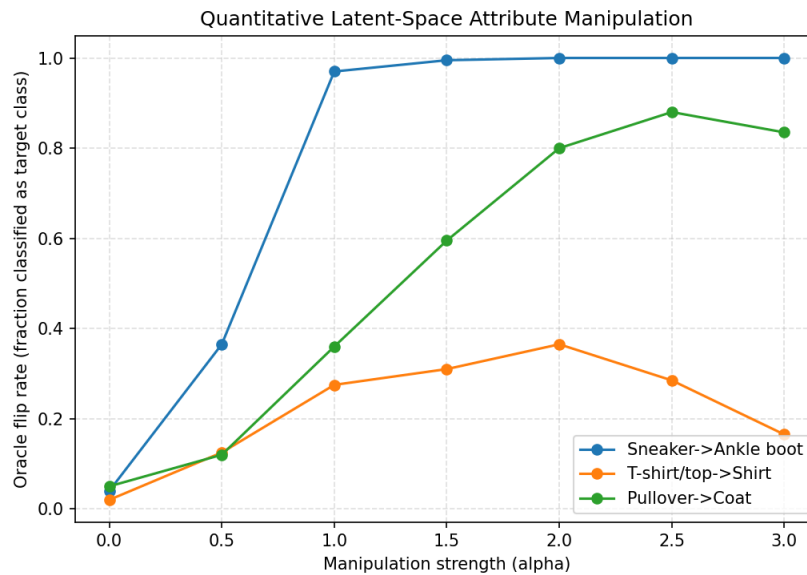


Figure 5. Oracle flip rate vs. manipulation strength α , one curve per transfer pair.

Table below reports the same data numerically:

Transfer Pair	$\alpha=0$	$\alpha=0.5$	$\alpha=1$	$\alpha=1.5$	$\alpha=2$	$\alpha=2.5$	$\alpha=3$
Sneaker $\rightarrow$ Ankle boot	4.0%	36.5%	97.0%	99.5%	100.0%	100.0%	100.0%
Pullover $\rightarrow$ Coat	5.0%	12.0%	36.0%	59.5%	80.0%	88.0%	83.5%
T-shirt/top $\rightarrow$ Shirt	2.0%	12.5%	27.5%	31.0%	36.5%	28.5%	16.5%

The results support the hypothesis, with an important qualification: success depends strongly on how visually distinct the source and target classes are in the underlying data, not just on whether a semantic label exists for the pair. Sneaker $\rightarrow$ Ankle boot reaches a 97–100% flip rate by $\alpha=1.0$ — both classes share the same viewing angle and differ mainly in ankle height, a feature the latent direction captures cleanly. Pullover $\rightarrow$ Coat rises more gradually to 83–88%, consistent with Figure 4 showing these classes as overlapping but not identical clusters. T-shirt/top $\rightarrow$ Shirt peaks at only 36.5% ($\alpha=2.0$) and then declines — in Figure 4 these two classes occupy almost the same latent region, so the mean-difference direction is short and noisy, and past $\alpha\approx 2$ the shifted latent likely leaves the manifold of realistic images altogether (consistent with the qualitative grids in Figure 6, where the T-shirt $\rightarrow$ Shirt sequence changes silhouette only subtly even at $\alpha=3.0$).

4.5 Qualitative Manipulation Grids

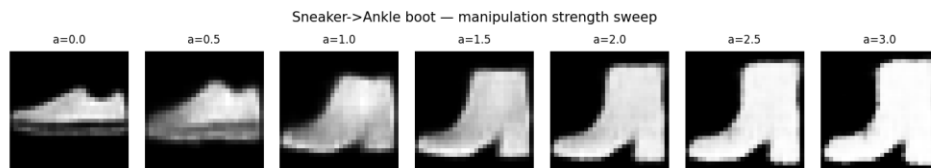


Figure 6a. Sneaker $\rightarrow$ Ankle boot: a clear, monotonic rise in ankle height as α increases, matching the near-100% flip rate.

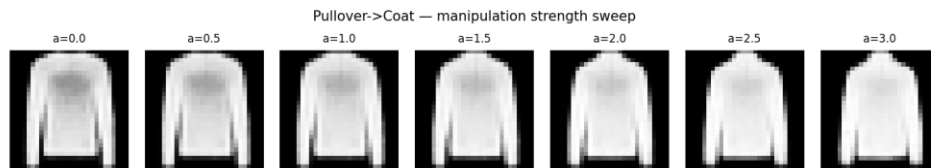


Figure 6b. Pullover $\rightarrow$ Coat: the garment silhouette gradually elongates and the collar area changes, consistent with the mid-range flip rate.

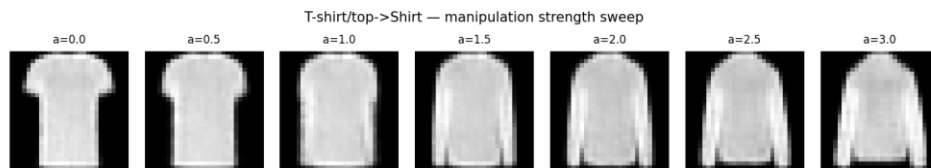


Figure 6c. T-shirt/top $\rightarrow$ Shirt: only a subtle sleeve-length change is visible even at high α , consistent with the low, non-monotonic flip rate — the decoded images plausibly drift off the data manifold rather than cleanly becoming a “Shirt.”

5. Discussion & Observation

1. Interpolating between two points in a VAE's latent space yields smooth, meaningful images; the same interpolation in a plain autoencoder often yields garbage. Why? What is the KL term doing to the geometry of the latent space to make interpolation work?

A plain autoencoder's encoder is free to scatter encodings anywhere in latent space, as long as the decoder can invert them — there is no pressure for the region between two valid encodings to itself decode to anything sensible, so the

space can have gaps, holes, or disconnected islands that an interpolated midpoint may land in. The KL term in the VAE loss penalizes the encoder's output distribution for departing from a standard normal prior, which pulls every encoding toward a shared, densely-packed, roughly convex region centered at the origin. Because every point in that region is encouraged to be a plausible sample under the prior — and the reconstruction loss simultaneously forces the decoder to make every such point decode to something realistic — the space between any two real encodings is also filled with points the decoder has been trained to render sensibly. The interpolation results in Figure 3 show exactly this: each intermediate frame is a coherent, if novel, garment shape rather than a blurry or nonsensical blend, which is the qualitative signature of the KL term successfully imposing a smooth, filled geometry on the latent space.

2. Suppose you replaced the Gaussian prior with something else (a mixture, or a uniform ball). How would sampling and the shape of the latent space change, and what might break?

Replacing the single Gaussian with a mixture-of-Gaussians prior (e.g. one component per class) would let the model carve the latent space into separate, potentially disjoint high-density regions instead of forcing everything into one shared blob — sampling would first pick a component and then a point within it, which could improve sample diversity and class-conditional generation but would also weaken the guarantee this project relies on for Section 4.4: if the T-shirt and Shirt clusters became genuinely separate rather than overlapping as in Figure 4, moving along a straight-line direction between their means might cross a low-density gap between components, and interpolation could reintroduce the same “garbage in between” failure mode a plain autoencoder has, just now between mixture components instead of everywhere. A uniform prior over a bounded ball, by contrast, would encourage encodings to spread out to fill the ball's volume more evenly (rather than concentrating near the origin as under a Gaussian); this could improve sample diversity near the boundary but removes the natural, densest-at-the-center sampling distribution that makes naïve ancestral sampling ($z \sim \text{prior}$) work well under a Gaussian, so decoded samples near the boundary — where the model may have seen fewer training encodings — could become less reliable.

6. Limitations & Next Steps

- The PCA projection in Figure 4 is a linear projection of a 20-D space onto 2 dimensions; classes that appear to overlap in PC1/PC2 may be more separated along other latent directions not shown here, so the manipulation results should be read as evidence about the specific mean-difference direction tested, not as a full characterization of class separability in the latent space.
- The oracle classifier itself is only 89.24% accurate, so some fraction of “non-flips” (especially in the T-shirt/top→Shirt pair, where the two classes are known to be visually similar even to a supervised classifier) may reflect oracle uncertainty rather than a genuine manipulation failure.
- Flip rate for T-shirt/top→Shirt peaks at $\alpha=2.0$ and then declines, suggesting the shifted latent leaves the region the decoder was trained on at high α ; a natural follow-up is to check reconstruction quality (e.g. via the oracle's confidence, or a separate realism check) as a function of α to confirm this.
- As the handbook's stretch goal suggests, sample quality was assessed only qualitatively (Figure 2); a follow-up could quantify it with FID against real Fashion-MNIST images.